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Redox equilibria of iron in low- to high-silica melts: A simple model and its applications to C-H-O-S degassing

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ABSTRACT

Delineating the redox variations of magmatic processes relies on quantitative models based on the redox equilibria of multi-valent elements, such as Fe and S, in the melt and/or minerals. Much effort has been made to quantify the redox equilibrium between ferrous and ferric iron (Fe^{2+} and Fe^{3+}) in silicate melts; however, existing Fe-redox models display large, systematic uncertainties for low- and high-SiO_2 melts. Here we present a new Fe-redox model that can predict the oxygen fugacity (fO_2) and Fe^{3+}/Fe^{2+} ratios for magmatic liquids with 0 – 79 wt% SiO_2 at variable temperatures and pressures. Our model calibration is based on an effective equilibrium constant using a comprehensive compilation of experimental data ($N = 1000$; 1 bar – 7 GPa; 800 – 1750 °C). Taking the recently published experiments ($N = 69$; 1 bar – 24 GPa; 900 – 2600 °C; $SiO_2 = 39 – 79$ wt%) as independent tests, we show that our model can be extrapolated to silicate melts at higher pressure and temperature. Applications of this model to representative melts demonstrate that adiabatic transport from the source to surface does not appear to vary the fO_2 of mid-ocean ridge basalt and rhyolite melts but can increase the fO_2 of alkali ocean island basalt (OIB) melts and peridotite magma ocean liquids by ~2 – 3 log units. Notably, the adiabatic fO_2 profile of alkali OIB indicates an auto-oxidation process controlled by Fe redox equilibria, decoupling the fO_2 of OIBs and their mantle sources. For an improved understanding of the redox evolution during magma degassing, we calibrate a new predictive model for $S^{6+}/\Sigma S$ ratios in silicate melts. With the newly developed Fe- and S-redox models, we then construct a new Magma And Gas Equilibrium Calculator (MAGEC) with improved user-settings that can simulate S degassing in dynamically evolving gas-melt coupled magmatic systems with a broad range of compositions. Using olivine-hosted melt inclusion data from silica-undersaturated lavas, we demonstrate that the new MAGEC has predictive capacities of redox equilibria and S distribution during magmatic degassing, making it a useful tool for a quantitative understanding of volatile behaviors in various magmatic systems relevant to planetary differentiation and crustal formation.

1. Introduction

Knowledge of the redox condition is essential to understand the chemical differentiation and dynamic evolution of the solid Earth. The oxygen fugacity (fO_2) describes the redox condition of a petrological system and is often normalized by mineral redox buffers (e.g., FMQ: Fayalite-Magnetite-Quartz: IW: Iron-Wüstite; and NNO: Ni-NiO) on a log scale, such as ΔFMQ ($= \log fO_2 - \log fO_2[FMQ]$). Many previous studies have demonstrated that the normalized fO_2 could change during crustal magmatic processes through degassing and crystallization (e.g., Moretti and Papale, 2004, Moussallam et al., 2016; Sun and Lee, 2022; Tang et al., 2020), and during large-scale planetary processes by core formation and plate tectonics (Cottrell et al., 2021; Evans, 2012; Righter et al., 2016; Stagno and Aulbach, 2021). To what extent crustal- and planetary-scale processes can affect the redox condition of Earth and rocky planets, however, is still largely uncertain. Redox equilibria of multi-valent elements (e.g., Fe and S) are key to this question in that they can not only enable inverse constraints on the fO_2 of natural rocks but also allow forward simulations of fO_2 variations due to magma degassing and crystallization.

As a major element with two oxidation states (Fe^{3+} and Fe^{2+}), Fe in the melt has been a primary interest in understanding the redox condition of magmatic systems. Much effort has been made previously to parameterize Fe redox equilibria in silicate melts as a function of fO_2, melt composition, temperature (T), and pressure (P) using experimental data (see references in Table 1). The most popular Fe-redox model for

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silicate melts was calibrated by Kress and Carmichael (1991) (denoted as KC91 hereafter) mainly for basaltic and andesitic melts at ~1000 – 1600 °C and 1 bar. The *P* effect in the KC91 model was adjusted using the compression properties derived from acoustic velocities. More recent models constrained the *P* effect using experimental measurements (Armstrong et al., 2019; Kuwahara et al., 2023; O'Neill et al., 2006; Zhang et al., 2017) or first-principles molecular dynamic simulations (Deng et al., 2020). Although the recent Fe-redox models are expected to provide better predictions, a systematic comparison of these models and a thorough assessment on their accuracies have not been conducted, leaving it unclear which model is suitable for a broader range of *P-T*-composition during crustal- and planetary-scale processes.

In this study, we first assess the previous Fe-redox models through (1) fO_2 predictions for melts from various geological processes and (2) a comprehensive compilation of experimental data from the literature. We demonstrate that the classic KC91 model and other models have large uncertainties for silicate melts distinct from typical basalt-andesite compositions or at high *P* conditions. Using the compiled experimental data, we then develop a new predictive model to describe the Fe redox equilibria in magmatic liquids. In addition, for better understanding the redox variations during magma degassing, we calibrate a new model for S redox equilibria in silicate melts. Integrating the new Fe- and S-redox models into the framework of Magma And Gas Equilibrium Calculator (MAGEC; Sun and Lee, 2022), we then construct a new MAGEC program to simulate C-H-O-S behaviors in chemically, dynamically evolving gas-melt coupled systems. Finally, we use melt inclusion data to demonstrate that the new MAGEC can better explain the redox evolution and degassing history of alkali melts particularly at sulfide saturation.

2. Assessment of previous Fe-redox models

2.1. Comparison of previous models

Considering four melts with fixed ferric to total iron ratios ($Fe^{3+}/\Sigma Fe$), we calculated their adiabatic fO_2 profiles (Fig. 1) for different potential temperatures (T_p) using the classic KC91 model and more recent models from O'Neill et al. (2006) (ON06), Armstrong et al. (2019) (A19), Deng et al. (2020) (D20), Hirschmann, (2022) (H22), and Kuwahara et al. (2023) (K23). The two models from Hirschmann (2022) were both considered because they used different Equations of State (EOS) from A19 and D20, respectively (denoted as H22A and H22D hereafter), to constrain the *P* effect. The selected melts include a mid-ocean ridge basalt (MORB; $Fe^{3+}/\Sigma Fe = 0.1$; $T_p = 1300$ °C), an alkali ocean-island basalt (OIB; $Fe^{3+}/\Sigma Fe = 0.3$; $T_p = 1500$ °C), a rhyolite ($Fe^{3+}/\Sigma Fe = 0.3$; $T_p = 900$ °C), and a peridotite liquid ($Fe^{3+}/\Sigma Fe = 0.05$; $T_p = 1800$ °C). Detailed compositions are listed in Table S1. For the

Table 1
The compilation of experimental data on Fe redox equilibria in magmatic liquids.

ID	Refs.	N[$]	*P* (GPa)[&]	*T* (°C)	logfO_2	$Fe^{3+}/\Sigma Fe$	SiO_2 (wt%)	Method[#]
1	Kennedy (1948)	7	atm	1200 – 1430	−0.68	0.71 – 0.86	51.12	W
2	Fudali (1965)	27	atm	1200	−8.50 to −0.68	0.12 – 0.88	47.49 – 61.48	W
3	Sack et al. (1980)	53	atm	1201 – 1330	−8.46 to −6.77	0.10– 0.25	39.50 – 76.10	W
4	Thornber et al. (1980)	44	atm	1196 – 1366	−8.10 to −6.02	0.12 – 0.24	42.04 – 59.85	W
5	Kilinc et al. (1983)	42	atm	1340 – 1454	−0.68	0.62 – 0.89	39.71 – 75.28	W
6	Kress and Carmichael (1988)	51	atm	1249 – 1635	−8.91 to −0.68	0.06 – 0.83	39.60 – 57.34	W
7	Kress and Carmichael (1989)	52	atm	1241 – 1635	−10.46 to −0.68	0.05 – 0.94	28.83 – 58.42	W
8	Kress and Carmichael (1991)	29	atm	996 – 1541	−0.68	0.62 – 0.99	29.29 – 70.79	W
9	Moore et al. (1995)	49	atm – 0.18	950 – 1550	−10.40 to −0.68	0.08 – 0.88	54.62 – 77.90	W
10	Kudo et al. (2000)	131	atm	1200 – 1300	−13.00 to −10.70	0.02 – 0.31	0.00 – 51.54	W
11	Gaillard et al. (2002)	4	0.2	800	−15.36 to −12.47	0.23 – 0.60	73.76 – 73.84	W
12	Gaillard et al. (2003)	22	atm – 0.2	1000 – 1210	−11.06 to −8.86	0.08 – 0.44	68.88 – 77.85	W
13	Jayasuriya et al. (2004)	28	atm	1409	−10.90 to −0.01	0.02–0.85	36.82 – 49.97	M
14	Partzsch et al. (2004)	26	atm	1132 – 1222	−11.34 to −6.72	0.07 – 0.40	48.43 – 49.99	M
15	Botcharnikov et al. (2005)	16	0.2	1200	−9.22 to −3.95	0.08 – 0.60	49.20 – 50.32	M
16	Wilke et al. (2005)	22	atm	1200 – 1350	−7.59 to −0.01	0.07 – 0.93	47.59 – 53.81	M,W
17	Mysen (2006)	9	atm	1300 – 1375	−8.00 to −0.68	0.09 – 0.81	38.30 – 46.84	M
18	O'Neill et al. (2006)	6	0.4 – 3	1400	−0.59 to 0.28	0.71 – 0.80	57.84	M
19	Henao and Itagaki (2007)	56	atm	1300 – 1400	−8.00 to −6.00	0.12 – 0.63	0.00 – 46.85	W
20	Schuessler et al. (2008)	16	0.05 – 0.5	1200 – 1250	−9.26 to −4.38	0.18 – 0.50	49.28 – 50.81	W
21	Cottrell et al. (2009)[†]	13	atm	1350	−8.72 to −2.20	0.08 – 0.58	49.70 – 50.89	M
22	Borisov and McCammon (2010)	22	atm	1398 – 1571	−0.68	0.58 – 0.77	48.15 – 75.08	M
23	Bell and Simon (2011)	4	0.13	950	−11.90 to −9.20	0.28 – 0.42	67.37 – 67.62	X
24	Borisov et al. (2013)	43	atm	1400 – 1552	−0.68	0.41 – 0.76	31.14 – 67.65	W
25	Borisov et al. (2015)	67	atm	1399 – 1551	−2.90 to −0.68	0.32 – 0.79	36.66 – 67.07	W
26	Cicconi et al. (2015)	15	atm	1250 – 1400	−9.77 to −0.68	0.06–0.91	48.68 – 51.03	X
27	Zhang et al. (2016)[†]	19	atm	1350	−8.63 to −0.68	0.07 – 0.75	58.15 – 60.28	M, X
28	Borisov et al. (2017)	53	atm	1450 – 1553	−0.68	0.42 – 0.77	36.44 – 65.82	W
29	Zhang et al. (2017)	19	atm – 7	1400 – 1750	−0.73 to 2.75	0.49 – 0.72	57.20 – 59.04	M
30	Berry et al. (2018)	10	atm	1400	−9.00 to 0.00	0.02 – 0.77	52.00	M
31	Borisov et al. (2018)	31	atm	1301 – 1500	−0.68	0.48 – 0.80	44.58 – 65.29	W
32	Davis and Cottrell (2018)	8	atm	1225	−9.89 to −5.79	0.08 – 0.39	55.56 – 57.42	X
33	Moussallam et al. (2019)	6	atm	1400	−7.30 to −4.27	0.08 – 0.34	44.85	M
34	Armstrong et al. (2019)	16	4 – 20	1400 – 2200	0.54 to 6.45	0.44 – 0.80	46.19 – 60.53	M
35	Blundy et al. (2020)	25	atm – 0.42	952 – 1400	−10.30 to −0.68	0.06 – 0.80	46.38 – 64.53	M, X
36	Sossi et al. (2020)	12	atm	1839 – 1924	−7.86 to 0.00	0.01 – 0.44	46.37 – 47.32	X
37	Matzen et al. (2022)	8	atm	1300 – 1500	−7.20 to −0.68	0.08 – 0.66	50.61 – 50.90	M
38	Kuwahara et al. (2023)	7	15 – 24	2300 – 2600	−1.99 to 0.29	0.13 – 0.20	39.37 – 50.08	X
39	Tiraboschi et al. (2023)	1	2	900	−6.90	0.79	79.13	M
Calibration data (ID # 1 – 33)		1000	atm – 7	800 – 1750	−15.36 to 2.75	0.02 – 0.99	0.00 – 77.90	W, M, X
Testing data (ID# 34 – 39)		69	atm – 24	900 – 2600	−10.30 to 6.45	0.01 – 0.80	39.37 – 79.13	M, X

[†] Calibration methods updated by Zhang et al. (2017).
[$] Number of experiments.
[&] atm – 1 bar.
[#] Analytical methods for Fe_2O_3: X – X-ray Absorption Near Edge Spectroscopy (XANES); M – Mössbauer; W – Wet chemistry.

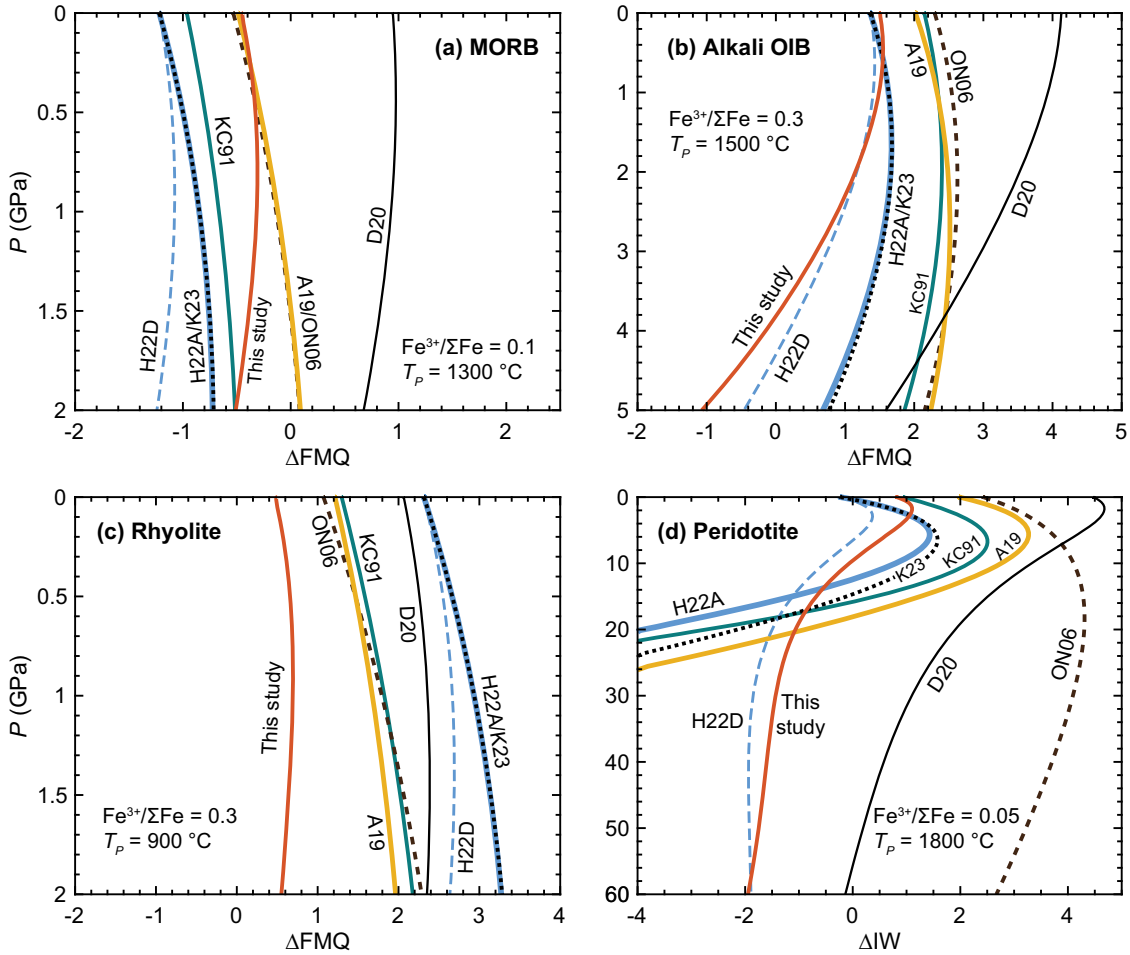

Fig. 1. Adiabatic redox profiles derived from Fe-redox models in the literature and this study for MORB (a), alkali OIB (b), rhyolite (c) and peridotite melts (d). Previous Fe-redox models include KC91 (Kress and Carmichael, 1991), ON06 (O'Neill et al., 2006), A19 (Armstrong et al., 2019), D20 (Deng et al., 2020), H22A/D (Hirschmann, 2022), and K23 (Kuwahara et al., 2023). See details of the melt compositions and potential temperatures (T_p) in Table S1.

peridotite liquid, we solved the adiabatic *P-T* path through numerical integration of the *P/T* gradient using the thermodynamic parameters from Thomas and Asimow (Thomas and Asimow, 2013). Assuming constant thermal expansion coefficient (α), density (ρ), and specific heat (c_p), we calculated the adiabatic *P-T* paths for MORB, alkali OIB, and rhyolite using a simple expression,

$$T = T_p \exp[r(P - 0.0001)] \quad (1)$$

where P is in GPa, T is in K, and r is the thermal expansion parameter ($r = \alpha/\rho/c_p$) set at 0.068 GPa^{-1}.

As shown in Fig. 1a–b, the fO_2 predictions of KC91 for MORB and alkali OIB are similar to those of ON06, A19, H22A, H22D, and K23 but are progressively smaller than those of D20 by up to ~2 orders of magnitude from high to low *P*. For the rhyolite (Fig. 1c), the KC91 predictions are comparable to those of ON06 and A19 but are ~1 order of magnitude smaller than those of D20, H22A, H22D, and K23 at low *P*. The calculated adiabatic fO_2 profiles for the peridotite liquid (Fig. 1d) show that the KC91 predictions are overall similar to those of A19, H22A, and K23 across a wide range of *P*, manifesting an apparent "redox transition" at ~5 – 10 GPa. Similar "redox transitions", however, appear at deeper depths (~15 GPa) in the predictions of ON06 but emerge at shallower depths (~2 GPa) in the predictions of D20 and H22D. The considerable discrepancies among the predictions of previous models indicate that large uncertainties are likely involved in the existing models at certain *P-T*-composition ranges. Therefore, it is necessary to systematically assess these models for a broad range of *P-T*-composition using a comprehensive compilation of experimental data.

2.2. Data compilation

We compiled high-*T* experiments on Fe redox equilibria in magmatic liquids from the literature according to the following criteria: (1) $Fe^{3+}/\Sigma Fe$ ratios of the quenched melts were determined with relative uncertainties <100 %; (2) the redox conditions were controlled at fixed fO_2 buffers (e.g., gas mixtures or mineral redox buffers) or monitored independently; (3) the experiments approached chemical equilibrium or showed no obvious signs of disequilibrium; (4) major elements were well-characterized for the quenched melts. The Fe speciation data determined by wet chemical methods, Mössbauer spectroscopy, and X-ray Absorption Near Edge Spectroscopy were all included in our compilation with the assumption that inter-laboratory discrepancies of these analytical methods play a secondary role in the variation of measured $Fe^{3+}/\Sigma Fe$ ratios. As crystals often have distinct Fe compositions from silicate melts, we excluded data from *bulk* analysis of experimental melts with quenched crystals. Quenched melts with high totals (> 102 wt%) were also excluded. Experiments for extremely alkali-rich systems (K_2O > 15 wt% or Na_2O > 15 wt%) were not included in our compilation because of potentially distinct charge-balancing effect on Fe^{3+} (Tangeman et al., 2001). In addition, we also excluded the two 23-GPa runs from Armstrong et al. (2019) and the four runs at 27 – 28 GPa from Kuwahara et al. (2023). The former show

extremely high $Fe^{3+}/\Sigma Fe$ ratios (= 0.95 – 0.96) in the quenched phase under reduced redox conditions (IW+0.8), while the latter contain small metal droplets that may have significantly modified the $Fe^{3+}/\Sigma Fe$ ratios of quenched phases.

After filtering, we obtained 1069 experiments from 39 studies as listed in Table 1 and detailed in Table S2. The compilation covers a wide range of $Fe^{3+}/\Sigma Fe$ ratios (= 0.01 – 0.99), redox conditions ($\log fO_2$ = −15.4 to 6.5; ΔIW = ∼−2 to 14), T (= 800 – 2600 °C), P (= 1 bar – 24 GPa), and melt compositions (e.g., SiO_2 = 0 – 79 wt%; ΣFeO = 0.6 – 92 wt%). Experiments at >7 GPa were only reported by two recent studies (Armstrong et al., 2019; Kuwahara et al., 2023) that were subject to extensive quench modifications and thus may have larger uncertainties in the chemical analyses. Melt compositions of the compiled experiments range from silica undersaturation to oversaturation with total alkali contents ($Na_2O + K_2O$) of up to ∼17 wt% (Fig. 2a), including rhyolites, basalts, peridotite liquids, and alkali-rich melts. The $Fe^{3+}/\Sigma Fe$ ratios are positively correlated with $\log fO_2$ (ΔIW = ∼−2 to 14; Fig. 2b) generally following a simple power-law relation, $\log(Fe^{3+}/Fe^{2+}) = 0.25\log fO_2 + c$, in which c is a constant. At a specific redox condition, however, the $Fe^{3+}/\Sigma Fe$ ratios also show significant variations (e.g., $Fe^{3+}/\Sigma Fe$ = ∼0.3 – 0.8 at IW+6; Fig. 2b), indicating the effects of P-T-composition on Fe speciation in the melt. Hence, without constraints on P-T-composition, the $Fe^{3+}/\Sigma Fe$ ratios of silicate melts are unlikely reliable proxies for melt fO_2.

2.3. Uncertainties of previous models

Using the compiled experimental data, we tested the classic KC91 model and six other models (ON06, A19, D20, H22A, H22D, and K23) calibrated for variable P-T-composition. To compare the model predictions, we calculated the Chi-Square value (χ^2) for each model,

$$\chi^2 = \sum_{j=1}^{N} \frac{(e_j - p_j)^2}{|p_j|} \tag{2}$$

where N (= 1064) is the total amount of data in our compilation; e_j and p_j denote the experimental and predicted $\log fO_2$ values for an experiment j; $|p_j|$ is the absolute value of the predicted $\log fO_2$. In general, a smaller χ^2 value indicates better predictions. The χ^2 values of the ON06 (χ^2 = 610) and A19 (χ^2 = 694) models are ∼3–4 times smaller than those of the KC91 (χ^2 = 3220) and D20 (χ^2 = 2525) models. The χ^2 values of the H22A (χ^2 = 920), H22D (χ^2 = 932), and K23 (χ^2 = 910) models are in the middle but appear to be very similar. Although the ON06 and A19 models have the lowest χ^2 values, their predictions still show significant deviations from the experimental $\log fO_2$ values (Fig. 3). We note that the uncertainties of $\log fO_2$ predictions from individual models vary systematically as a function of SiO_2 in the melt. Such systematic errors cannot be simply attributed to inter-laboratory discrepancies of different analytical methods but instead indicate strong effects of melt compositions (e.g., SiO_2) overlooked by previous models.

The classic KC91 model reproduces the experimental $\log fO_2$ values for basaltic to andesitic melts (SiO_2 = ∼50 – 60 wt%) to within ∼1 log unit, including some of the high-P melts from Armstrong et al. (2019) (ID# 34 in Table 1), but overestimates the $\log fO_2$ values for silica-rich or silica-poor melts by up to ∼8 log units (Fig. 3a). The ON06 model predicts the $\log fO_2$ for andesitic and rhyolitic melts to within ∼1 log unit but appears to systematically underestimate the $\log fO_2$ for silica-poor melts by up to ∼4 log units (Fig. 3b). Notably, the ON06 model overestimates the $\log fO_2$ for high-P melts (ID# 34 and 38 in Table 1) and low-P, high-Ti melts (ID# 24 in Table 1) by up to ∼9 log units. The A19 model shows smaller predictive uncertainties than the KC91 model, particularly for the silica-rich melts, but still overestimates the $\log fO_2$ values for silica-poor melts by up to ∼5 log units (Fig. 3c). The D20 predictions (Fig. 3d) appear to be better than those of the A19 and ON06 models but show an overall large uncertainty (χ^2 = 2525) as they are systematically greater than the experimental values. The D20 model also overestimates the experimental $\log fO_2$ for peridotite liquids (ID# 36 and 38 in Table 1) by ∼2 – 6 log units. Similar to the KC91 model, the H22A and H22D models also show better predictions for melts with intermediate SiO_2 but systematically overestimates the $\log fO_2$ for melts with lower (< ∼50 wt%) or higher (> ∼60 wt%) SiO_2 by up to ∼4 log units (Fig. 3e, f). Large uncertainties of up to ∼3 log units also appear in the predictions of the H22A and H22D models for hydrous basalts (ID# 35 in Table 1). We note that the H22A and H22D models reproduce the $\log fO_2$ of low-P peridotite liquids from Sossi et al. (2020) (ID# 36 in Table 1) to within ±1 log unit but generate larger uncertainties for the high-P peridotite liquids from Kuwahara et al. (2023) (ID# 38 in Table 1). As the K23 model was modified from the H22A model, it yields predictions (Fig. 3h) very similar to H22A and also appear to overestimate the $\log fO_2$ for some of their own experiments by ∼2 – 4 log units. Noting the systematic variations of prediction uncertainties from previous models, in the following section we use the compiled data to develop a new Fe-redox model for a broad range of fO_2-P-T-composition space.

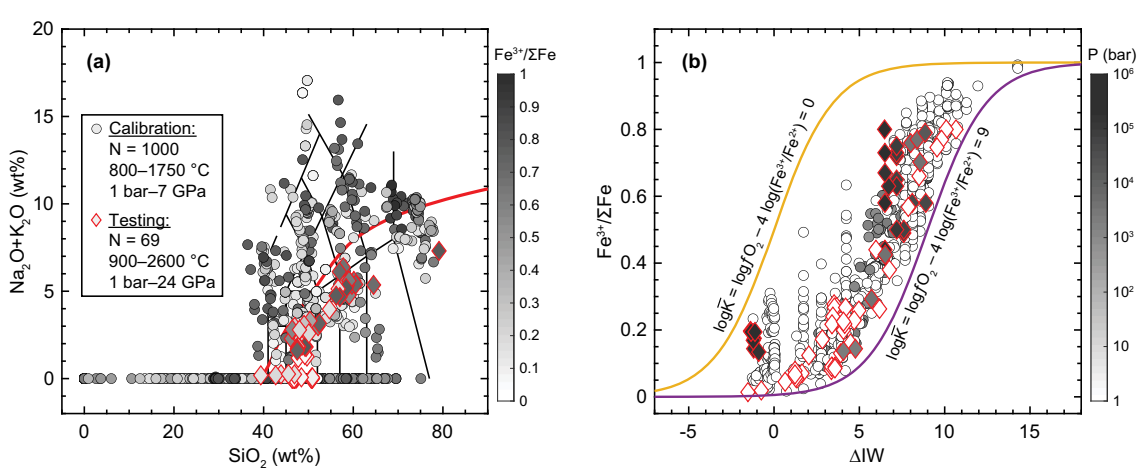

Fig. 2. Compositional and redox variations of the compiled experiments. Plot (a) is the Total Alkali and Silica (TAS) diagram showing the melt compositions, in which marker colors indicate the $Fe^{3+}/\Sigma Fe$ ratios. Plot (b) shows the $Fe^{3+}/\Sigma Fe$ ratios as a function of redox conditions (ΔIW) with markers color-coded by the experimental pressure (P). Solid curves in (b) show two values of the effective equilibrium constant (cf. Eq. (5)) to bracket the compiled experiments. Circle and diamond markers denote two subsets of the compiled experimental data used for the model calibration and independent tests, respectively.

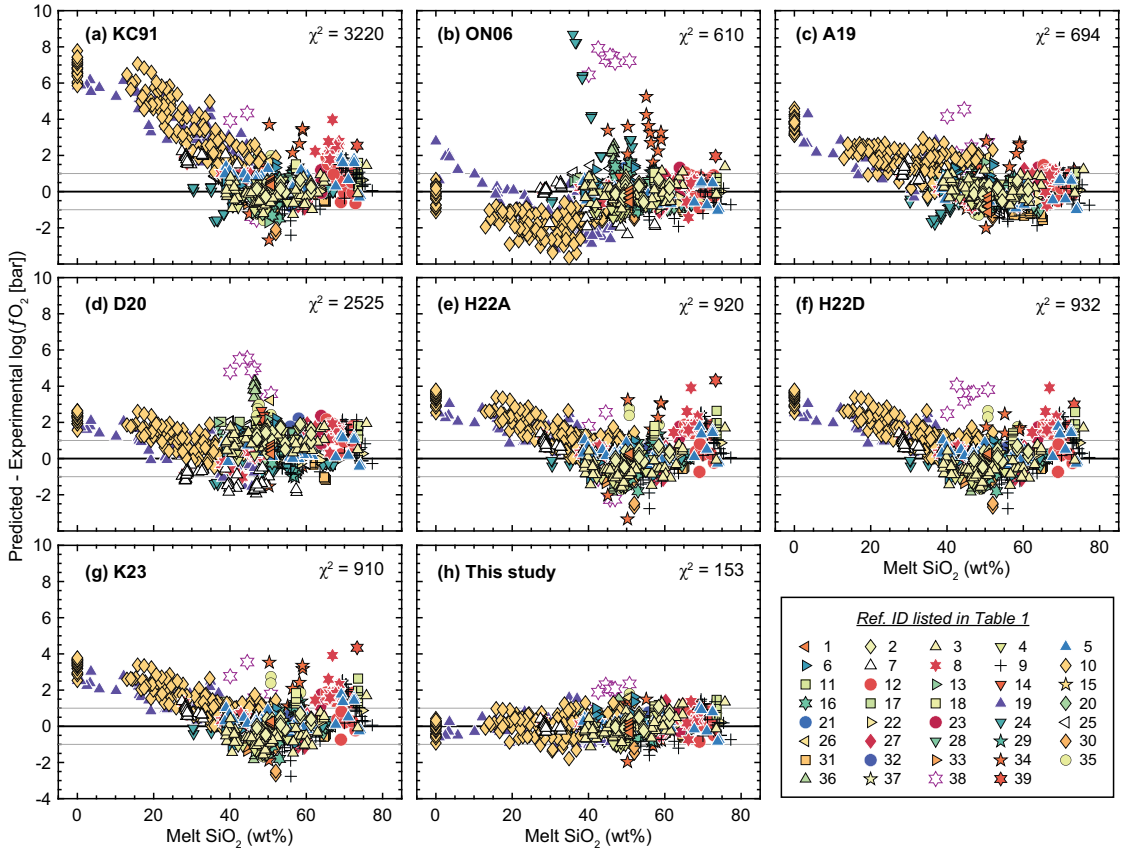

Fig. 3. Uncertainties of the predicted $\log fO_2$ derived from Fe-redox models in the literature and this study for the compiled experiments. See the caption of Fig. 1 for references of previous Fe-redox models. Table 1 lists the references (ID# 1–39 in Table 1) for the compiled experiments. The Chi-Square value (χ^2) is defined in Eq. (2) to compare the relative uncertainties of model predictions. Note the systematically larger uncertainties of previous models for low- and high-SiO_2 melts.

3. A new Fe-redox model for magmatic liquids

3.1. Thermodynamic basis

Ferric-ferrous redox equilibrium in silicate melts could be described using simple oxides or ionic ligands. It has been argued that the ionic approach can better characterize the compositional effect by considering the redox reaction with stable Fe–O polyhedrons, such as Fe^{2+}–FeO_4^{5-} (Moretti and Neuville, 2021). However, implementing this approach for high-P conditions is challenging due to the lack of constraints on stable ionic ligands at elevated P. As the oxide approach only deals with two simple components (FeO and $FeO_{1.5}$) regardless of P-T-composition variations, it has been used more widely in the literature. Following the simple-oxide approach, the ferric-ferrous redox equilibrium in a magmatic liquid can be evaluated through the reaction,

$$FeO + \frac{1}{4}O_2 = FeO_{1.5} \tag{3}$$

The equilibrium constant of this reaction leads to a general expression to describe the Fe^{3+}/Fe^{2+} ratio of the magmatic liquid as a function of fO_2-P-T-composition,

$$\begin{aligned} \log\left(\frac{X_{FeO_{1.5}}}{X_{FeO}}\right) &= \frac{1}{4}\log fO_2 - \log\left(\frac{\gamma_{FeO_{1.5}}}{\gamma_{FeO}}\right) - \frac{\Delta G(T,P_0)}{RT\ln(10)} - \int_{P_0}^{P}\frac{\Delta V}{RT\ln(10)}dP \\ \Delta G(T,P_0) &= \underbrace{\left(\Delta H(T_0,P_0) + \int_{T_0}^{T}\Delta C_P dT\right)}_{\Delta H(T,P_0)} - T \times \underbrace{\left(\Delta S(T_0,P_0) + \int_{T_0}^{T}\Delta C_p d\ln T\right)}_{\Delta S(T,P_0)} \end{aligned} \tag{4}$$

where X_i and γ_i denotes the molar fraction and activity coefficient, respectively, of component i (FeO or $FeO_{1.5}$) in the melt; T_0 and P_0 denote the reference condition; ΔG, ΔH, ΔS, ΔC_P, and ΔV are the changes of Gibbs free energy, enthalpy, entropy, heat capacity, and volume, respectively, for the reaction (Eq. (3)) with pure oxide liquids (FeO and $FeO_{1.5}$) at specific T and P; and R is the gas constant.

The explicit expression of Eq. (4) is determined by the formulations of $\gamma_{FeO1.5}/\gamma_{FeO}$, ΔG, and ΔV, which are largely empirical due to the non-ideal behaviors of FeO and $FeO_{1.5}$ in multi-component melts and due to the lack of direct measurements on the thermodynamic properties of pure $FeO_{1.5}$ liquid. Because the $\gamma_{FeO1.5}/\gamma_{FeO}$ ratio for multi-component melts varies with the melt composition, including the Fe^{3+}/Fe^{2+} ratio, the apparent exponent (n) in the power-law relation, $\log(Fe^{3+}/Fe^{2+}) \sim n\log fO_2$, could deviate from the expected stoichiometry coefficient ($= 1/4$) in the redox reaction (Eq. (3)). As outlined initially in Sack et al. (1981), two approaches may handle this issue, and both have been implemented in the literature. Specifically, one could either consider the coefficient of $\log fO_2$ as an adjustable parameter (e.g., KC91, H22A/D, and K23) or assume a regular solution model for FeO and $FeO_{1.5}$ (e.g., ON06, A19, and D20). In existing models, ΔG was often approximated as

a simple function of T, $\Delta G = c_0 + c_1T + c_2T\ln T$, where c_0, c_1, and c_2 denote the coefficients. This expression can be derived from Eq. (4) using constant ΔC_P (e.g., KC91, H22A/D, and K23), and the "$T\ln T$" term may be further removed by assuming negligible ΔC_P (e.g., ON06 and A19). However, additional T terms have also been introduced to account for the T dependence of ΔG (e.g., D20). On the other hand, ΔV (= $V_{FeO1.5}$ – V_{FeO}) was evaluated previously through EOS with parameters constrained either by high-P experiments (e.g., ON06 and A19) or by first-principle molecular dynamic simulations (e.g., D20). The systematic errors in the predictions of existing models (Fig. 3) indicate that the aforementioned treatments need to be improved to develop a better Fe-redox model.

3.2. Calibration strategy

We define an effective equilibrium constant ($\overline{K} = fO_2\ X^4_{FeO}\ /X^4_{FeO_{1.5}}$) to maintain the theoretic stoichiometry relation between fO_2 and $FeO_{1.5}$/FeO, which also enables us to explore the non-ideal behaviors of $FeO_{1.5}$ and FeO in the melt. This approach is distinct from all previous studies that treated Fe^{3+}/Fe^{2+} and fO_2 as two independent variables for model calibrations (e.g., KC91, ON06, A19, H22, D20, H22, and K23). According to Eq. (4), $\log\overline{K}$ can be described as

$$\log\overline{K} = \log fO_2 - 4\log\left(\frac{X_{FeO_{1.5}}}{X_{FeO}}\right) = 4\log\left(\frac{\gamma_{FeO_{1.5}}}{\gamma_{FeO}}\right) + \frac{4\Delta G(T,P_0)}{RT\ln(10)} + \frac{4}{\ln(10)}\int_{P_0}^{P}\frac{\Delta V}{RT}dP \quad (5)$$

Considering the T dependence of ΔC_P, we assume ΔG as a combination of variable T terms (e.g., $c_0 + \sum_{i=1}^{N} c_iT^n + c_{N+1}T\ln T$; $n = -1$, 0.5, 1, 1.5, 2, …). To quantify the compositional effect, we allow $\log(\gamma_{FeO1.5}/\gamma_{FeO})$ to vary with $\log(Fe^{3+}/Fe^{2+})$, ln(ΣFeO), SiO_2, and other compositional parameters, assuming that $\gamma_{FeO1.5}$ and γ_{FeO} are dependent on the concentrations of $FeO_{1.5}$, FeO, and other major oxides in the melt. As direct experimental measurements on ΔV at high P are not available, we constrain the P effect using the 4th-order Birch-Murnaghan EOS of Deng et al. (2020) derived from first-principles molecular dynamic simulations for simple silicate melts (i.e., $Mg_{14}Fe_2Si_{16}O_{48}$ and $Mg_{14}Fe_2Si_{16}O_{49}$) at P up to 140 GPa. Deng et al. (2020) obtained this EOS by assuming a linear combination of oxide molar volumes in the melts (i.e., ideal mixing). Noting this assumption may slightly over-simplify the coordination environments of Fe^{3+} and Fe^{2+} in multi-component melts, we add an adjusting factor (β) to empirically constrain ΔV (i.e., $\Delta V = \beta\ \Delta V_{D20}$).

Additional considerations are taken to make better use of the compiled experimental data. Because the compiled data only cover P from 1 bar to 24 GPa (Table 1), it is possible that any models calibrated using this dataset may not be reliable for extrapolation to deep mantle conditions. To mitigate this issue, we divide our compilation into two datasets for model calibration and testing, respectively. The former includes 1000 data with P = 1 bar – 7 GPa and T = 800 – 1750 °C from earlier literature (ID# = 1–33; in Table 1), whereas the latter has 64 data from more recent studies (ID# = 34–39 in Table 1) covering a broader range of P (= 1 bar – 24 GPa) and T (= 900 – 2600 °C). The calibration dataset (N = 1000), however, has an uneven data distribution with 897 at 1 bar and 103 at 0.05 – 7 GPa, which could potentially undermine the predictive accuracy of the new model at high P. Accordingly, we take a two-step approach in the calibration by (1) first developing a 1-bar Fe-redox model and (2) then adding minimal correction terms to fit the high-P data (P = 0.05 – 7 GPa). At each step, we conduct least-squares regression analyses of the calibration data to identify key parameters of P-T-composition through an extensive search of various permutations. Regardless of the choice of parameters, the new model should reproduce the recent experimental data from the testing dataset (N = 69; P = 1 bar – 24 GPa; ID# =34–39 in Table 1) with a reasonable accuracy. Although the testing dataset includes fewer experiments, it provides essential, independent validation of the new model at P-T conditions beyond the calibration range.

3.3. A new Fe-redox model for magmatic liquids

Through the calibration methods outlined above, we obtained the following empirical expression to describe $\log\overline{K}$ at 1 bar (= P_0) as a function of T and composition:

$$\begin{aligned}\log\overline{K}(T,P_0) &= a_0X^{0.5}_{\Sigma FeO}\log\left(\frac{X_{FeO_{1.5}}}{X_{FeO}}\right) + \Omega(T) + \Phi(X)\\ \Omega(T) &= a_1 + a_2T^{1.5} + a_3\ln T\\ \Phi(X) &= a_4\ln(X_{\Sigma FeO}) + a_5X^{0.5}_{\Sigma FeO} + a_6X^3_{SiO_2} + a_7X_{AlO_{1.5}} + a_8X_{TiO_2}\\ &\quad + a_9X_{CaO} + a_{10}X_{MgO} + (a_{11} + a_{12}X_{\Sigma FeO})\left(X_{NaO_{0.5}} + X_{KO_{0.5}}\right)\end{aligned} \quad (6)$$

where a_0, a_1, … a_{12} are the fitting coefficients from least-squares regression analyses of the 1-bar experimental data (N = 897) in the calibration dataset; X_i denotes the mole fraction of a single-cation oxide (i) in the melt; and Ω and Φ are intermediate variables. Ω is closely related to $\Delta G/RT$ in Eq. (5), indicating a T dependence of ΔC_P ($\sim c_0 + c_1T^{1.5}$). Using the high-P experimental data (N = 103; P = 0.05 – 7 GPa) from the calibration dataset, we then quantified the P effect through,

$$\begin{aligned}\log\overline{K}(T,P) &= \log\overline{K}(T,P_0) + h\cdot\Gamma(P,T)\\ \Gamma(P,T) &= \int_{P_0}^{P}\frac{\Delta V_{D20}}{RT}dP\end{aligned} \quad (7)$$

where h is an adjusting coefficient (= $4\beta/\ln(10)$) determined by least-squares regression, and Γ can be evaluated using the EOS of Deng et al. (2020) or our parameterization in Table S3. Table 2 lists the fitting coefficients and their uncertainties in Eqs. (6)–(7).

A new Fe-redox model can then be derived from Eqs. (6)–(7) to predict the $\log fO_2$ at specific T and P using the Fe^{3+}/Fe^{2+} ratio of a magmatic liquid,

$$\log fO_2 = \left(4 + a_0X^{0.5}_{\Sigma FeO}\right)\log\left(\frac{X_{FeO_{1.5}}}{X_{FeO}}\right) + \Omega(T) + \Phi(X) + h\cdot\Gamma(P,T) \quad (8)$$

If the $\log fO_2$ is provided, one could estimate the Fe^{3+}/Fe^{2+} ratio of a magmatic liquid for the P-T conditions of interest through the following expression.

$$\log\left(\frac{X_{FeO_{1.5}}}{X_{FeO}}\right) = \frac{\log fO_2 - \Omega(T) - \Phi(X) - h\cdot\Gamma(P,T)}{4 + a_0X^{0.5}_{\Sigma FeO}} \quad (9)$$

An Excel® worksheet is provided in the supplementary materials (Table S4) for interested users to implement this model. With the inputs of P, T,

Table 2
Fitting coefficients in Eqs. (6)–(7).

	Value	Standard error
a_0	2.1479	0.0689
a_1	−230.2593	22.5093
a_2	-1.8557×10^{-4}	0.3405×10^{-4}
a_3	34.3293	3.3464
a_4	1.4138	0.0617
a_5	−17.3040	0.4718
a_6	−10.1820	0.5411
a_7	−6.7463	0.5033
a_8	−7.3886	0.7740
a_9	−14.5430	0.3911
a_{10}	−9.9776	0.5245
a_{11}	−16.1506	0.7075
a_{12}	−37.5572	6.7315
h	2.1410	0.1319

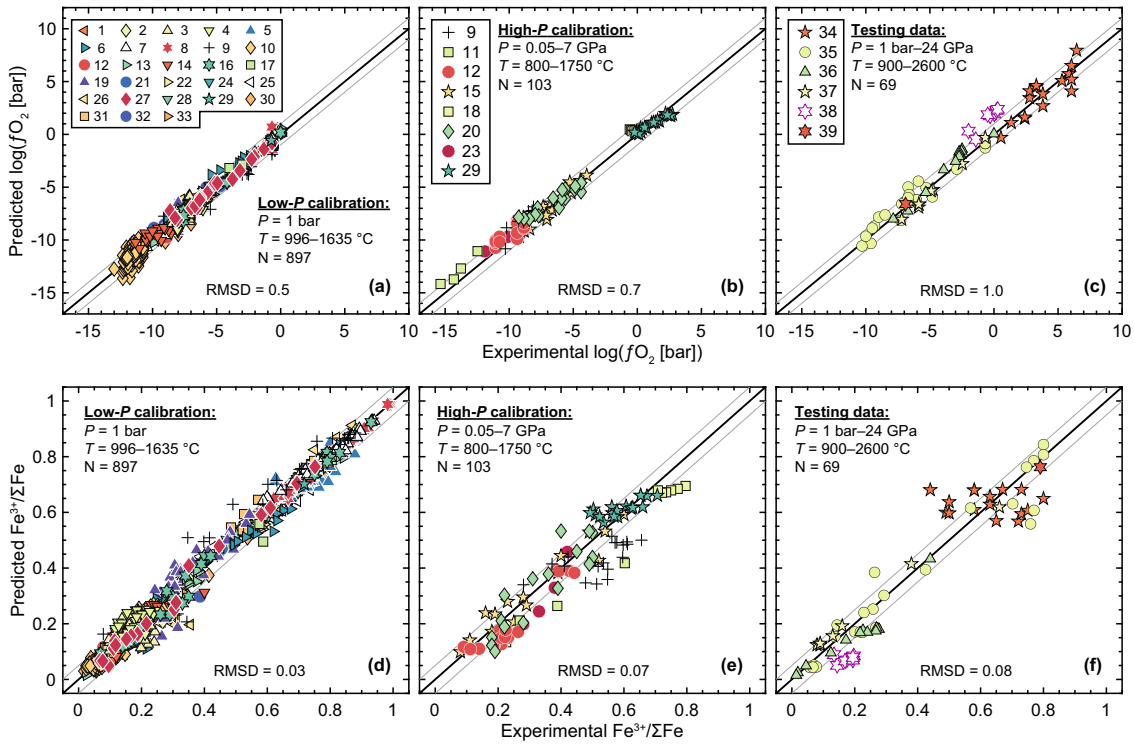

Fig. 4. Comparisons in logfO_2 (a-c) and $Fe^{3+}/\Sigma Fe$ (d-f) between experimental and predicted values for the calibration (1 bar and 0.05–7 GPa) and testing datasets (1 bar–24 GPa). The predicted values are calculated using the new Fe-redox model from this study. The Root Mean Square Deviations (RMSD) indicate the uncertainties of our model predictions. Larger uncertainties are evident for high-*P* data (ID# = 34 and 38 in Table 1) with extensive quench modifications.

and melt compositions, this worksheet automates the calculations of logfO_2 and $Fe^{3+}/\Sigma Fe$ ratios for single or multiple samples through a macro button.

Applying this new model to the compiled experimental data, we compared the predicted logfO_2 and $Fe^{3+}/\Sigma Fe$ ratios with the experimental values in Fig. 4. To measure the prediction uncertainty, we computed the Root Mean Square Deviation (RMSD) for two subsets of calibration data (1-bar vs. high-*P*; ID# = 1–33; in Table 1) and the testing data (ID# = 34–39 in Table 1). The predicted logfO_2 for the 1-bar (Fig. 4a) and high-*P* calibration data (Fig. 4b) agree well with the experimental values with RMSD values of 0.5 and 0.7, respectively. Although the testing data were excluded from the calibration, our model can reproduce their logfO_2 values with a RMSD of 1.0 (Fig. 4c); this larger uncertainty is mainly due to the high-*P* data (> ~20 GPa) that underwent extensive quench modifications, especially those from Kuwahara et al. (2023). As noted by the author correction, Kuwahara et al.'s experimental data likely underwent more significant quenched modifications than they thought.

The $Fe^{3+}/\Sigma Fe$ ratios from our model prediction are in excellent agreement with the reported values from the 1-bar experiments, indicating a RMSD of 0.03 (Fig. 4d). The high-*P* data (0.05 – 24 GPa) also show large uncertainties (RMSD = 0.07 – 0.08) in the predicted $Fe^{3+}/\Sigma Fe$ ratios (Fig. 4e, f), which is due to extensive quench modifications. In general, our new model shows a high performance in predicting the logfO_2 and $Fe^{3+}/\Sigma Fe$ ratios of the testing data (Figs. 3, 4) and therefore is arguably accurate and suitable for magmatic systems with a broad range of *P-T*-composition. Importantly, the χ^2 value (= 153) of our model for the logfO_2 of compiled experiments is significantly smaller than those (χ^2 = 688 – 3220) of previous models (Fig. 3f), further demonstrating the improved accuracy and applicability of our model. Considering the limited high-*P* data (> 20 GPa), we caution that our Fe-redox model needs to be tested and improved by future high-*P* experiments without substantial quench modifications.

3.4. Effects of pressure, temperature, and melt composition

For a magmatic liquid at specific *P* and *T*, Eqs. (8)–(9) indicate a linear correlation between logfO_2 and log(Fe^{3+}/Fe^{2+}), overall similar to previous Fe-redox models (e.g., KC91, ON06, A19, D20, H22A/D, and K23). However, in the space of logfO_2 vs. log(Fe^{3+}/Fe^{2+}), the slope of this line defined by our model increases with ΣFeO in the melt, which is distinct from the constant slopes in previous models. The intercept of this line determined by Γ(*T, P*), Ω(*T*) and Φ(*X*) in our model includes non-linear terms of *P, T*, and composition. These non-linear terms, however, make it difficult to build intuitive understanding on to what extent *P, T*, or composition can affect the relation between logfO_2 and log(Fe^{3+}/Fe^{2+}). To demonstrate the effects of *P* and *T*, we calculated the $Fe^{3+}/\Sigma Fe$ ratios for a range of logfO_2 (= −15 to 10) at three different *P* (= 1 bar, 5 GPa, or 15 GPa) and two choices of *T* (= 1300 °C and 1800 °C) using the MORB composition in Table S1. As shown in Fig. 5a, the Fe^{3+}/Fe^{2+} redox transitions are generally across ~16 log units. Raising *T* from 1300 °C to 1800 °C at a given *P* can increase fO_2 by ~5 orders of magnitude for a fixed $Fe^{3+}/\Sigma Fe$ ratio, while isothermally compressing the melt with a fixed $Fe^{3+}/\Sigma Fe$ ratio from 1 bar to 15 GPa can increase fO_2 by ~4 orders of magnitude. The FMQ-normalized logfO_2 (i.e., ΔFMQ) does not follow such large extents of variations because the logfO_2 of mineral redox buffers also vary with *P* and *T* but in different manners. For instance, at a fixed $Fe^{3+}/\Sigma Fe$ ratio (= 0.3) and 2 GPa, ΔFMQ of the MORB melt increases by ~1.5 log units as *T* increases from 900 °C to ~1400 °C but starts to decrease as *T* goes above ~1400 °C (Fig. 5b).

The compositional effect on the melt fO_2 is highlighted in Fig. 5b by showing ΔFMQ as a function of *T* for the peridotite, alkali OIB, and rhyolite liquids (cf. Table S1) at 2 GPa for a fixed $Fe^{3+}/\Sigma Fe$ ratio (= 0.3). The ΔFMQ values of the peridotite liquid are nearly identical to those of MORB because of their similarities in key compositional parameters, such as SiO_2 (45 *vs.* 49 wt%) and ΣFeO (8.05 *vs.* 7.97 wt%). With lower SiO_2 (= 43 wt%) and higher ΣFeO (= 12 wt%), the alkali OIB melt has

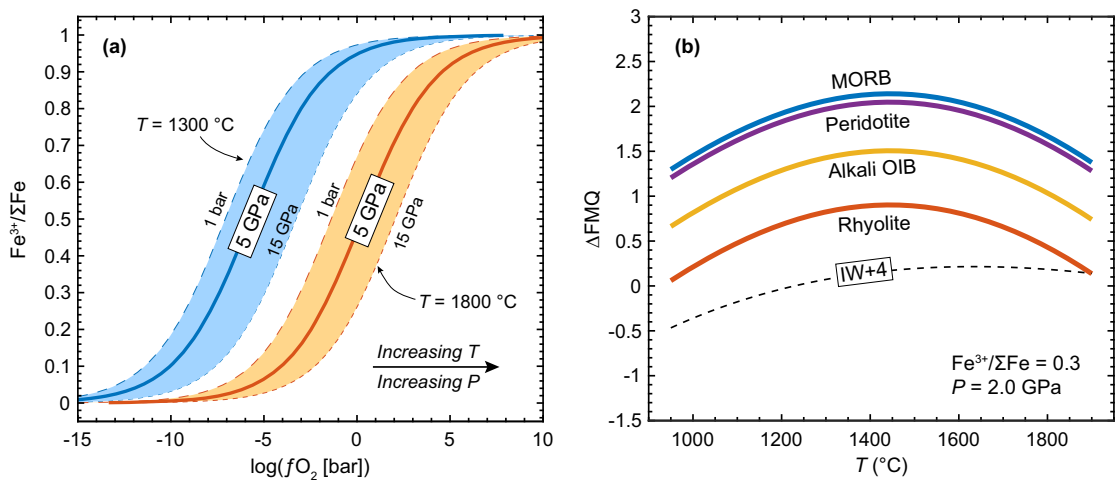

Fig. 5. Effects of temperature (T), pressure (P), and melt composition on the redox responses of the melt $Fe^{3+}/\Sigma Fe$ ratios. Plot (a) shows the correlation between the $Fe^{3+}/\Sigma Fe$ ratio and $\log fO_2$ for three selected P (1 bar, 5 GPa, and 15 GPa) at two different T (1300 °C and 1800 °C). Plot (b) shows the redox variations (ΔFMQ) of MORB, peridotite, alkali OIB, and rhyolite melts as a function of T at 2 GPa for a fixed $Fe^{3+}/\Sigma Fe$ ratio (= 0.3). Dashed curve in (b) displays the ΔFMQ variation at IW+4.

lower ΔFMQ values than the MORB melt by ~0.5 log units. Because the rhyolite melt is more silica-rich and Fe-poor (SiO_2 = 77 wt%; ΣFeO = 0.7 wt%) than the other melts, it shows lower ΔFMQ values than the MORB melt by ~1 log unit, indicating a significant effect of melt compositions on the relation between the fO_2 and $Fe^{3+}/\Sigma Fe$ ratio. Considering their strong influences on the fO_2 and $Fe^{3+}/\Sigma Fe$ ratio, P-T-composition must be considered and well characterized to better determine the magmatic fO_2 and the redox condition of magma source regions using the melt $Fe^{3+}/\Sigma Fe$ ratio, and vice versa.

3.5. Magma redox variations during adiabatic ascent

Melt transport from the source to (near) surface is associated with changes in P and T, which is often approximated as adiabatic decompression assuming no significant heat loss. Prior to extensive differentiation and degassing, the redox evolution of ascending melts is controlled by Fe redox equilibrium due to the combined effect of P and T and thus can be simulated using Eq. (8) for a given adiabatic P-T path. Using the new Fe-redox model (Eq. (8)), we calculated the adiabatic fO_2 profiles (Fig. 1) for four representative melts (i.e., MORB, alkali OIB, rhyolite, and peridotite liquid). The melt compositions, $Fe^{3+}/\Sigma Fe$ ratios, and T_p are taken from Table S1. For MORB, our model predictions recover those of A19 and ON06 at P < ~1 GPa but closely match the results of KC91, H22A, and K23 at ~2 GPa. For the alkali OIB, our model predictions are similar to those of H22D, and both are distinct from the predictions of other models at elevated P. Notably, our model shows significant discrepancies from all previous ones by ~1 – 3 log units in the predicted adiabatic fO_2 profiles for the rhyolite melt across the selected P range (1 bar to 2 GPa). For the peridotite liquid, our model shows similar predictions as H22D but yields higher $\log fO_2$ values at P < 50 GPa by up to ~1 log unit.

Several intriguing points can be made from the adiabatic fO_2 profiles derived by our new Fe-redox model. First, the calculated ΔFMQ values for the MORB (Fig. 1a) and rhyolite melts (Fig. 1c) appear to be nearly constant at shallower depths (< 2 GPa). This indicates that these melts, if subject to minimal degassing (see Section 4 for degassing), can record the redox conditions of their source regions. Second, the predicted ΔFMQ for the alkali OIB can increase by ~2.5 log units as it segregates adiabatically from 5 GPa to ~1 GPa (Fig. 1b), suggesting a stronger P/T effect at deeper depths and higher adiabatic T. This implies that a deeper depth of melt generation yields a greater extent of redox decoupling between the source region and the ascending melt. Because primary alkali OIB melts are generated from plume mantle sources at ~4 – 5 GPa (Sun and Dasgupta, 2023), i.e., ~3 GPa deeper than those of MORBs, OIBs upon ascending to shallower depths are expected to be more oxidized than MORBs by ~1 – 2 log units even if their mantle sources share similar redox conditions (e.g., ΔFMQ = ~ −0.5). This finding suggests that the deeper depth of melt generation could be an alternative explanation or make additional contributions to the relatively oxidized redox conditions of OIBs observed by previous studies (e.g., Cottrell et al., 2021; Moussallam et al., 2019). Finally, the adiabatic fO_2 profile of the peridotite liquid reaches Fe-metal saturation at ~60 GPa (i.e., ~IW-2). If this depth marks the last equilibration P between the core and magma ocean along the adiabatic path at T_p = 1800 °C, our new Fe-redox model implies that Earth's core formation would place the $Fe^{3+}/\Sigma Fe$ ratio of the magma ocean at ~0.05 with a surface fO_2 at ~IW+1.

4. Implications for magma degassing simulations

During magma degassing, Fe redox plays a critical role in controlling the dissolved volatile species (e.g., S^{2-} and S^{6+}) in the melt and redox equilibria of C-H-O-S species in the gas. Several models have been constructed to simulate volcanic degassing of C-H-O-S volatiles (e.g., Burgisser et al., 2015; Ding et al., 2023; Hughes et al., 2023; Moretti et al., 2003; Sun and Lee, 2022; Witham et al., 2012). Because the existing models use the KC91 model to constrain Fe redox equilibria, they are not suitable for low- and high-SiO_2 magmas due to the intrinsic errors of the KC91 model (cf. Fig. 3a). In addition, most previous models could not handle the complexities of sulfide/sulfate saturation in conjunction with S redox equilibria and degassing, except for the MAGEC program of Sun and Lee (2022). As sulfide globules are common in olivine-hosted melt inclusions coexisting with vapor bubbles (e.g., Hartley et al., 2017; Kamenetsky and Zelenski, 2020), it is expected that sulfide saturation could be concurrent with volatile degassing. Because of the multi-valence states of S in the gas (2-, 0, and 4+) and melt (2- and 6+), saturation of sulfide (or sulfate) can buffer S^{2-} (or S^{6+}) in the melt and alter the total redox budget (or total oxygen) and volatile distribution in the gas-melt system. However, S redox in the earlier MAGEC was coupled with Fe^{3+}/Fe^{2+} using the model of Nash et al. (2019), in which the melt Fe^{3+}/Fe^{2+} ratio is determined by the KC91 model. To better understand C-H-O-S degassing, the MAGEC program needs to be reformed by incorporating our new Fe redox model and a more accurate S redox model.

4.1. Sulfur redox equilibria in silicate melts

We calibrated a new S redox model using 82 experimental data from the literature (Botcharnikov et al., 2011; Jugo et al., 2010; Klimm et al., 2012; Nash et al., 2019). These S redox data were measured for a broad range of melt compositions (e.g., $SiO_2 = 45 - 77$ wt%; $Na_2O = 0 - 21$ wt %) at $T = 850 - 1300$ °C, $P = 0.001 - 2$ kbar, and $\log fO_2 = -13.1$ to -5.7. Considering a simple S redox reaction (i.e., $S^{2-} + 2O_2 = SO_4^{2-}$), we obtained the following expression to describe the S^{6+}/S^{2-} ratios in silicate melts,

$$\log\left(\frac{S^{6+}}{S^{2-}}\right) = 2\Delta IW + 6.37(\pm 0.73) - \frac{23891(\pm 1185)}{T} + \frac{10.65(\pm 1.02)X_{SiO_2}X_{NaO_{0.5}}}{T} \times 10^4 + \frac{9.12(\pm 0.85)X_{SiO_2}X_{KO_{0.5}}}{T} \times 10^4 - 36.43(\pm 4.24)\left(X_{NaO_{0.5}} + X_{KO_{0.5}} - 4X_{PO_{2.5}} - X_{AlO_{1.5}}/6\right) \quad (10)$$

where numbers with uncertainties in paratheses are the fitting coefficients from least-squares regression analyses, and ΔIW is calculated using the model of Hirschmann (2021). Fig. 6a shows that Eq. (10) reproduces the experimental data to well within a factor of two. The χ^2 value (= 10) of our model is significantly smaller than those ($\chi^2 = 88 - 266$) of previous models (Boulliung and Wood, 2023; Jugo et al., 2010; Nash et al., 2019; O'Neill and Mavrogenes, 2022), indicating a substantial improvement of predictive accuracy. For interested users to implement this new S redox model, we added Eq. (10) to the Excel® worksheet in the supplementary materials (Table S4) that could automate calculations of $S^{6+}/\Sigma S$ ratios with the inputs of *P*-*T*-composition and redox conditions (e.g., $Fe^{3+}/\Sigma Fe$, $\log fO_2$, ΔIW, or ΔFMQ).

Our new S redox model indicates strong effects of *T* and composition on S redox equilibria in silicate melts. The *P* effect does not appear to be evident in Eq. (10) but is embedded in the IW buffer. To illustrate the *T*-composition effects, we calculated the $S^{6+}/\Sigma S$ ratios of four representative melts (MORB, alkali OIB, rhyolite, and peridotite) for ΔFMQ = −2 to +4 at 2 kbar and two choices of *T* (900 and 1300 °C). Fig. 6b shows that the S^{6+}/S^{2-} redox transitions extend across ~2 log units, which are ~8 times smaller than the Fe^{3+}/Fe^{2+} redox transitions (Fig. 5a). Increasing *T* shifts the S redox transition towards lower fO_2, indicating that higher *T* can stabilize S^{6+} in the melt for a fixed fO_2. At constant *T*, the S redox transitions also shift towards lower fO_2 from peridotite to rhyolite melts, but the extent of shifting decreases at higher *T*. Although the rhyolite melt appear to stabilize more S^{6+} than mafic and ultramafic melts at the same *T* and fO_2, this may not be the case in natural systems due to the competing effects between *T* and composition. To test the combined effects of *T* and composition, we calculated the $S^{6+}/\Sigma S$ ratios for these representative melts at relevant liquidus *T* (cf. Table S1). Fig. 6c shows that the S redox transition shifts towards lower fO_2 from the rhyolite melt at 900 °C to peridotite liquid at 1800 °C because the *T* effect overwhelms the compositional effect. Notably, natural rhyolite melts could be more oxidized than MORB melts even if they share identical $S^{6+}/\Sigma S$ ratios. Therefore, caution must be taken when estimating melt fO_2 using the S^{6+}/S^{2-} ratio.

4.2. A new MAGEC program for magma degassing

Incorporating our Fe and S redox models, we built a new MAGEC program, based on the earlier version in Sun and Lee (2022), for improved constraints on redox equilibria in the gas-melt system with C-H-O-S volatiles. The new MAGEC offers six options (e.g., $\log fO_2$, $Fe^{3+}/\Sigma Fe$, $S^{6+}/\Sigma S$, ΔIW, ΔFMQ, or ΔNNO) to set the initial fO_2 in the input file ("input" sheet). Note that the total oxygen derived from any of these redox options is *P*-dependent. The new MAGEC allows users to control the total oxygen in the input file through a "reference *P*", which will be set at the initial degassing *P* if left blank. The new MAGEC also includes user-friendly settings in the input file that enable users to choose: (1) alternative options for Fe/S redox equilibria, volatile solubility, sulfide capacity, and S^{2-} contents in the melt at sulfide saturation (i.e., SCSS), (2) opting out the treatment for saturation of graphite, sulfide, and/or sulfate, (3) oxygen-closed or oxygen-buffered systems, (4) *T*-*P* paths with the parameter *r* (= 0 for isotherms; Eq. (1)), and (5) ideal or real gas behaviors of individual vapor species. The MAGEC program and simulation examples (see below) are provided in the supplementary materials.

Using the new MAGEC program, we simulated closed-system degassing of alkaline melt inclusions from El Hierro, Canary Islands reported by Longpré et al. (2017). These melt inclusions are CO_2- and S-rich, while the hosting basanite lavas contain abundant sulfide globules, indicating sulfide saturation during degassing. The averaged major oxide compositions of the melt inclusions were used in the degassing simulation assuming negligible crystallization during degassing. According to Longpré et al. (2017), we set the undegassed melt with 3 wt% H_2O, 3 wt% CO_2, and a $S^{6+}/\Sigma S$ ratio of 0.7. Given the abundant sulfide globules in the hosting rocks, we assumed 1 wt% S in the bulk magmatic system as it should be significantly greater than the measured maximum value in melt inclusions (~0.5 wt% S). Applying the liquid thermometer of Sun and Dasgupta (2020), we obtained the liquidus *T* (~1110 °C) of these melt inclusions at 1 kbar, corresponding to T_p of ~1100 °C. As this

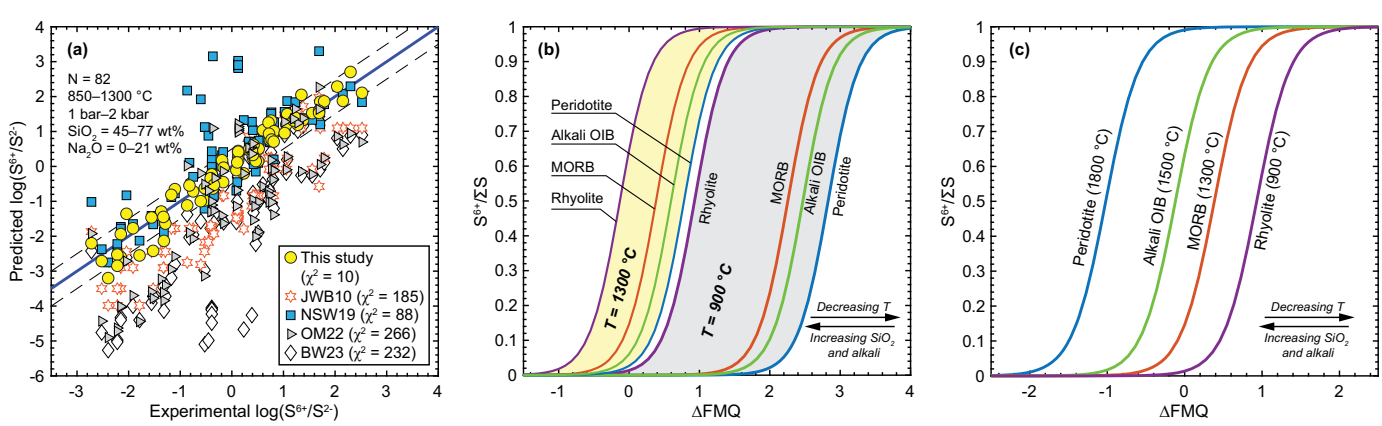

Fig. 6. A new S-redox model for predicting $S^{6+}/\Sigma S$ in silicate melts. Plot (a) shows the comparison between model predictions and experimental values of $\log(S^{6+}/S^{2-})$. Plots (b-c) show the variations of the model predicted $S^{6+}/\Sigma S$ ratios as a function of ΔFMQ for peridotite, alkali OIB, MORB, and rhyolite melts at hypothetic or relevant liquidus *T*. Previous S-redox models in (a) include JWB10 (Jugo et al., 2010), NSW19 (Nash et al., 2019), OM22 (O'Neill and Mavrogenes, 2022), and BW23 (Boulliung and Wood, 2023). Note that the two temperatures in (b) are not intended to match the petrogenesis of individual melts but are selected only to illustrate the *T* effect.

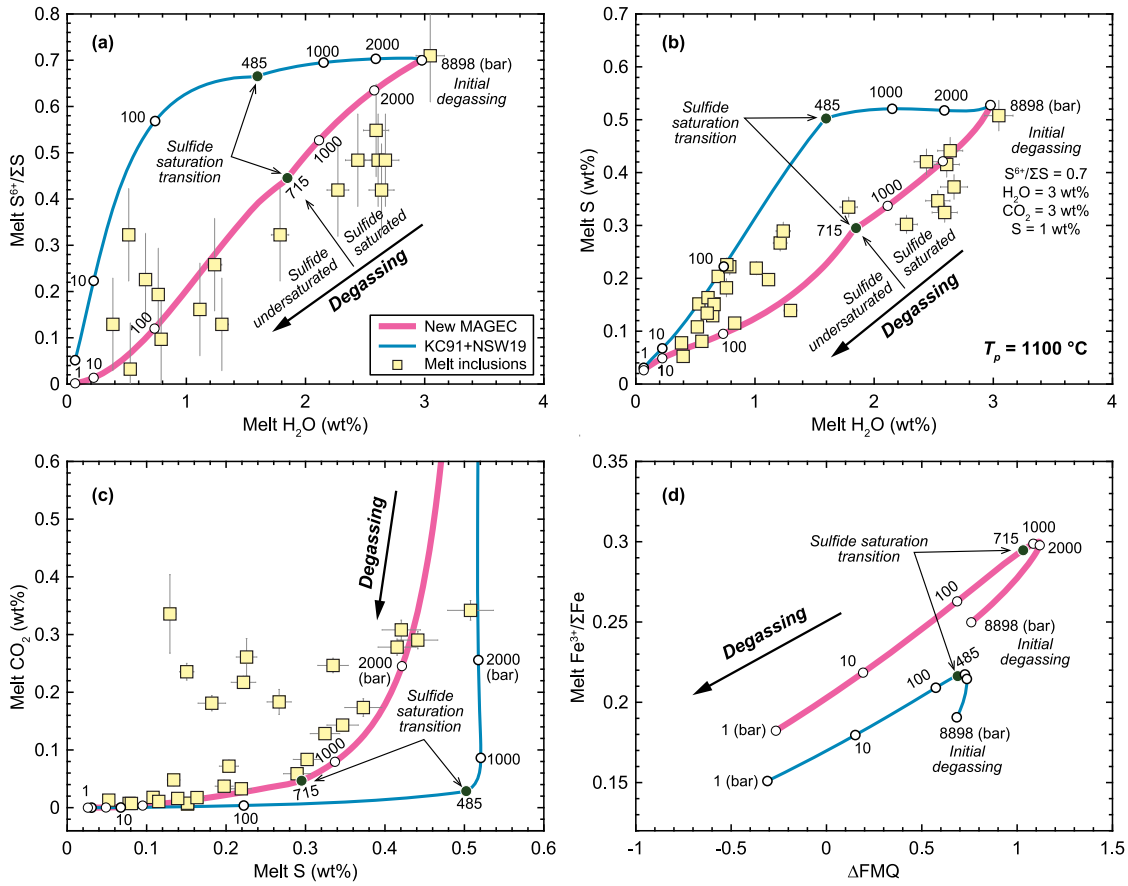

Fig. 7. MAGEC simulations of C-H-O-S degassing from a volatile-rich alkali basalt melt. Plots (a-c) show the predicted volatile compositions of the melt in comparison with those of olivine-hosted melt inclusions (yellow markers) reported by Longpré et al. (2017). Plot (d) displays the correlation between $Fe^{3+}/\Sigma Fe$ and ΔFMQ from model predictions. Blue and red curves indicate the simulation results using the Fe- and S-redox models either from previous studies (Kress and Carmichael, 1991; Nash et al., 2019) and this work (Eqs. (9)–(10)). Hollow circles labeled with numbers denote the degassing *P*, while filled circles indicate the transition from sulfide saturation to sulfide undersaturation.

T_p is significantly smaller than that (T_p = ~1490 °C) of the primary alkaline melts from Canary Islands (Sun and Dasgupta, 2023), these melt inclusions are likely derived from extensive differentiation and perhaps degassing of the primary melt in the deep crust.

Two sets of MAGEC simulations were performed using the same input settings on volatile solubility but different choices of Fe-S redox models from (1) this study and (2) KC91 and Nash et al. (2019). Both simulations predict the initial degassing at 8898 bar under sulfide saturation (Fig. 7), in which the initial gases are dominated by CO_2 for its low solubility in the melt at elevated *P*. The early degassing (e.g., > 2 kbar) can remove a large fraction of CO_2 (e.g., > 85 %) from the melt but only reduce H_2O in the melt by a small fraction (e.g., <~15 %). The choices of Fe-S redox models do not appear to affect the degassing trajectories of H_2O and CO_2 but yield distinct results in S degassing and melt fO_2. Sulfide saturation appears at 8898 – 715 bar in the first simulation but extends across 8898 – 485 bar in the second simulation. During degassing at sulfide saturation, the first simulation yields a slight increase in fO_2 by ~0.5 log units, where the second one generates nearly constant fO_2. When sulfide becomes undersaturated, both simulations predict a decrease of fO_2 (i.e., ~1.5 log units) induced by shallow degassing. The $S^{6+}/\Sigma S$ ratios show a monotonic decrease during degassing in the first simulation but only begin to decrease upon sulfide undersaturation in the second simulation. Importantly, the first simulation reproduces the variations of H_2O, S, and CO_2 in the melt inclusions as well as their redox history recorded by $S^{6+}/\Sigma S$, whereas the second simulation significantly overestimates the $S^{6+}/\Sigma S$ ratios and S contents in the melt. In addition, the predicted $Fe^{3+}/\Sigma Fe$ ratios (= ~0.25 – 0.30) of less-degassed melts from the first simulation are significantly greater than those from the second simulation ($Fe^{3+}/\Sigma Fe$ = ~0.19 – 0.22). The former are consistent with the recent measurements of $Fe^{3+}/\Sigma Fe$ (= ~0.27 – 0.30) by Moussallam et al. (2019), which independently confirms the improved accuracy of our new Fe-S redox models.

5. Conclusion

Using a comprehensive compilation of experimental data, we calibrate a new Fe redox model to quantify the relation between $\log fO_2$ and $Fe^{3+}/\Sigma Fe$ for magmatic liquids with variable compositions. Our new Fe redox model demonstrates significantly improved accuracy over the existing ones, particularly for low- and high-SiO_2 melts. This model can also be independently tested by recently published experimental data for rhyolite, andesite, basalt, and peridotite liquids at 1 bar to ~20 GPa. Assuming $Fe^{3+}/\Sigma Fe$ as the redox buffer for adiabatically ascending magmas without degassing, our Fe-redox model shows that (1) MORB and rhyolite melts maintain nearly constant fO_2 and thus could record the fO_2 of their sources; (2) alkali OIBs near surface become more oxidized than their mantle sources by ~2 log units, indicating that the depth of melt generation could serve as an alternative to mantle redox heterogeneity for the apparently high fO_2 of OIBs; and (3) the terrestrial magma ocean likely has a redox profile with increasing fO_2 of ~3 log units from the deep interior to surface. In addition, we construct a new S-redox model for silicate melts, which indicates a strong competing effect between *T* and composition on S redox equilibria. Combining our Fe-

and S-redox models, we develop a new MAGEC program to model the redox evolution and C-H-O-S degassing of gas-melt coupled magmatic systems. With user-friendly settings, the new MAGEC is suitable to simulate volatile degassing from low- to high-SiO_2 melts with variable *T*-*P*-fO_2 and compositions.

CRediT authorship contribution statement

Chenguang Sun: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Lijing Yao:** Writing – review & editing, Visualization, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All data supporting the findings of this study can be found in the supplementary material and the cited references.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.epsl.2024.118742.

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